

SEDAR 75 Gulf of Mexico Gray snapper

Recommendations for considerations of future research are provided below and do not indicate any particular order of priority.

Recreational Landings and Discards data

Further develop best practices for correcting for prominent peaks and troughs in the earlier part of the time series where uncertainty is high and catch/discard estimates are driven by few but influential intercept records.

Develop estimates of uncertainty around Headboat discard estimates.

Age and length composition

Quantify and evaluate appropriate modeling and weighting procedures of length and age compositions to ensure age and length composition inputs are representative of the segment of the population being modeled.

Improve sampling programs for shore based recreational fishing modes to obtain length and age data for that mode.

Selectivity and catchability of the commercial fleets

Further investigate and quantify changes in selectivity/catchability through time to improve fit to the discards and length compositions in recent years.

Continue data collection from observer programs or electronic monitoring programs.

Selectivity and retention of the recreational fleets

Further investigate and quantify changes in selectivity/catchability through time to improve fit to the length compositions across the time series.

Landings and Discards

Explore approaches for assigning uncertainty estimates to commercial landings and revisit estimation of historic landings.

Further investigate best practices for converting historical recreational landings from numbers to weight.

Recreational CPUE indices

Additional research is needed to investigate if assumptions are appropriate across full time series (e.g., targeting, trip length, effects of various regulations on gray snapper as well on other species i.e. red snapper).

Natural mortality

Explore ways to better reflect uncertainty about the mortality at age vector.